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Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – PERIODIC TEST 10
Chapter 13: Statistics

Time: 60 Minutes
MM: 30

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (6 × 1 = 6 Marks)

Q1. The median of a frequency distribution is found graphically using: [1]

(A) Histogram (B) Frequency polygon (C) Ogives (D) Pie chart

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) Ogives

Justification: Ogives are cumulative frequency curves ('less than' and 'more than'). The point of intersection of both ogives, or the abscissa of the point corresponding to $N/2$ on the ogive, gives the median of the distribution.

Q2. If the mean of 6 observations is 12 and each observation is divided by 2, then the new mean will be: [1]

(A) 12 (B) 6 (C) 24 (D) 3

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 6

Justification: If each observation x_i is divided by a non-zero constant $k = 2$, the new mean is:

$$\bar{x}_{\text{new}} = \frac{\bar{x}}{2} = \frac{12}{2} = \mathbf{6}.$$

New Mean = **6**

Q3. In a continuous frequency distribution, the modal class is $40 - 60$. If $l = 40, f_1 = 20, f_0 = 12, f_2 = 11, h = 20$, the mode is: [1]

(A) 48.42 (B) 49.41 (C) 50.25 (D) 47.85

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 49.41

Justification:

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h = 40 + \left(\frac{20 - 12}{2(20) - 12 - 11}\right) \times 20$$

$$\text{Mode} = 40 + \left(\frac{8}{40 - 23}\right) \times 20 = 40 + \frac{160}{17} = 40 + 9.4117... \approx \mathbf{49.41}.$$

$$\boxed{\text{Mode} = \mathbf{49.41}}$$

Q4. The cumulative frequency of the last class interval in a cumulative frequency table is equal to: [1]

(A) the class mark of the last class (B) the frequency of the first class

(B) the total frequency of the distribution (D) the range of the distribution

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) the total frequency of the distribution

Justification: The cumulative frequency of the final class interval aggregates all frequencies across the entire dataset, which by definition equals $N = \sum f_i$ (the total frequency).

Q5. If the mean of first n odd natural numbers is 15, then n is equal to: [1]

(A) 15 (B) 30 (C) 225 (D) 14

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 15

Justification: Sum of first n odd natural numbers $= n^2$.

$$\text{Mean} = \frac{n^2}{n} = n.$$

Given Mean $= 15 \implies \boxed{n = \mathbf{15}}$.

Q6. **Assertion (A):** If the mean of five numbers is 20, and one number is excluded, their mean becomes 18, then the excluded number is 28.
Reason (R): Excluded value is computed by subtracting the new total sum from the original total sum. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: Original sum $= 5 \times 20 = 100$.
New sum after excluding one observation $= 4 \times 18 = 72$.

$$\text{Excluded Number} = \text{Original Sum} - \text{New Sum} = 100 - 72 = 28.$$

Both statements are true and Reason (R) correctly explains Assertion (A).

Q7. Find the arithmetic mean of the following discrete frequency distribution: [2]

x	10	20	30	40
f	3	2	4	1

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Product Summation [1 Mark]

$$\sum f_i = 3 + 2 + 4 + 1 = 10.$$

$$\sum f_i x_i = (10 \times 3) + (20 \times 2) + (30 \times 4) + (40 \times 1) = 30 + 40 + 120 + 40 = 230.$$

Step 2: Mean Calculation [1 Mark]

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{230}{10} = \mathbf{23}.$$

$$\boxed{\text{Arithmetic Mean } (\bar{x}) = \mathbf{23}}$$

Q8. If the mode of a frequency distribution is 24 and the mean is 27, find its median using empirical relation. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Quoting the Empirical Relationship [1 Mark]

$$\text{Mode} = 3\text{Median} - 2\text{Mean}$$

Step 2: Substitution & Calculation [1 Mark]

Given Mode = 24 and Mean = 27:

$$24 = 3\text{Median} - 2(27) \implies 24 = 3\text{Median} - 54$$

$$3\text{Median} = 24 + 54 = 78 \implies \text{Median} = \frac{78}{3} = \mathbf{26}.$$

$$\boxed{\text{Median} = \mathbf{26}}$$

Q9. Convert the following 'more than' distribution into regular class intervals and calculate the mean: [3]

Production	≥ 0	≥ 10	≥ 20	≥ 30	≥ 40
Farms	50	46	38	20	8

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Conversion to Standard Intervals & Products [1.5 Marks]

Class Interval	Frequency (f_i)	Class Mark (x_i)	$f_i x_i$
$0-10$	$50-46=4$	5	20
$10-20$	$46-38=8$	15	120
$20-30$	$38-20=18$	25	450
$30-40$	$20-8=12$	35	420
$40-50$	8	45	360
Total	$\sum f_i = 50$		$\sum f_i x_i = 1370$

Step 2: Formula Declaration & Substitution [1 Mark]

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{1370}{50} = \frac{137}{5} = \mathbf{27.4}$$

Step 3: Final Boxed Result [0.5 Mark]

$$\boxed{\text{Mean Production} = \mathbf{27.4}}$$

Q10. Find the modal lifetime of 225 electrical components: [3]

Lifetime (hours)	$0-20$	$20-40$	$40-60$	$60-80$	$80-100$	$100-120$
Components	10	35	52	61	38	29

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Modal Class & Parameter Identification [1 Mark]

The maximum frequency is 61, which corresponds to the class **$60-80$**.
$\therefore$ Modal Class $= 60-80$.
Lower limit $(l) = 60$, class width $(h) = 20$.
$f_1 = 61$ (modal frequency), $f_0 = 52$ (preceding frequency), $f_2 = 38$ (succeeding frequency).

Step 2: Mode Formula & Substitution [1 Mark]

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

$$\text{Mode} = 60 + \left(\frac{61-52}{2(61)-52-38}\right) \times 20 = 60 + \left(\frac{9}{122-90}\right) \times 20 = 60 + \left(\frac{9}{32}\right) \times 20$$

Step 3: Calculation & Final Result [1 Mark]

$$\text{Mode} = 60 + \frac{180}{32} = 60 + 5.625 = \mathbf{65.625} \text{ hours} \quad \left(\text{or } 65\frac{5}{8} \text{ hours}\right)$$

$$\boxed{\text{Modal Lifetime} = \mathbf{65.625} \text{ hours} \quad \left(\text{or } \mathbf{65\frac{5}{8}} \text{ hours}\right)}$$

SECTION D: Long Answer Type

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Q11. The lengths of 40 leaves of a plant are measured correct to the nearest millimetre. Convert the data into continuous classes and calculate the median length: [5]

Length (mm)	118 – 126	127 – 135	136 – 144	145 – 153	154 – 162	163 – 171	172 – 180
Leaves	3	5	9	12	5	4	2

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Continuous Class Interval Conversion Table [2 Marks]

$$\text{Adjustment factor} = \frac{127 - 126}{2} = 0.5.$$

Given Class (mm)	**Continuous Class (mm)**	**Frequency (f_i)**	**Cum. Freq. (cf)**
118 – 126	117.5 – 126.5	3	3
127 – 135	126.5 – 135.5	5	8
136 – 144	135.5 – 144.5	9	17
145 – 153	**144.5 – 153.5**	**12**	**29**
154 – 162	153.5 – 162.5	5	34
163 – 171	162.5 – 171.5	4	38
172 – 180	171.5 – 180.5	2	40
Total		$N = 40$	

Step 2: Median Class & Parameters [1 Mark]

$N = 40 \implies \dfrac{N}{2} = 20.$

Cumulative frequency just greater than 20 is 29, corresponding to continuous class **144.5 – 153.5**.

$\therefore$ Median Class = 144.5 – 153.5.

Lower limit (l) = 144.5, $cf = 17$ (preceding class cf), $f = 12$, class width (h) = 9.

Step 3: Median Formula & Calculation [2 Marks]

$$\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f}\right) \times h = 144.5 + \left(\frac{20 - 17}{12}\right) \times 9 = 144.5 + \left(\frac{3}{12}\right) \times 9$$

$$\text{Median} = 144.5 + \frac{9}{4} = 144.5 + 2.25 = \mathbf{146.75} \text{ mm}.$$

$$\boxed{\text{Median Length of Leaves} = \mathbf{146.75} \text{ mm}}$$

Q12. Find the missing frequencies x and y in the following distribution table, given that the median is 32 and the total frequency is 100: [5]

Class	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60
Frequency	10	x	25	30	y	10

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► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Cumulative Frequency Table & Equation (i) [1.5 Marks]

Class Interval	Frequency (f_i)	Cumulative Frequency (cf)
$0-10$	10	10
$10-20$	x	$10+x$
$20-30$	25	$35+x$
$30-40$	30	$65+x$
$40-50$	y	$65+x+y$
$50-60$	10	$75+x+y$
Total	$N=100$	

Given total frequency $N=100$:

$$75+x+y=100 \implies x+y=100-75 \implies x+y=25 \quad \text{— (i)}$$

Step 2: Median Class Identification & Parameters [1 Mark]

Given Median $=32$, which lies in the class interval **30 – 40**.

$\therefore$ Median Class $=30-40$.

Lower limit $(l)=30$, $f=30$, $h=10$, $\dfrac{N}{2}=50$, $cf=35+x$ (preceding cf).

Step 3: Solving for x via Median Formula [1.5 Marks]

$$\text{Median}=l+\left(\frac{\frac{N}{2}-cf}{f}\right)\times h$$

$$32=30+\left(\frac{50-(35+x)}{30}\right)\times 10$$

$$2=\frac{15-x}{3} \implies 6=15-x \implies x=15-6=\mathbf{9}.$$

Step 4: Finding y & Final Result [1 Mark]

Substituting $x=9$ into Equation (i):

$$9+y=25 \implies y=25-9=\mathbf{16}.$$

$$\boxed{x=\mathbf{9}, \quad y=\mathbf{16}}$$

SECTION E: Case Study Based Question

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Q13. CASE STUDY: COMMERCIAL LOGISTICS FLEET FUEL EFFICIENCY BENCHMARKING [4 Marks]

A freight logistics enterprise tracked the operational fuel mileage (km per litre) of 60 heavy-duty cargo trucks operating across national supply corridors.

Mileage (km/l)	$10-12$	$12-14$	$14-16$	$16-18$	$18-20$
Truck Count	7	12	18	13	10

(i) Calculate the arithmetic mean fuel efficiency of the commercial fleet. [2]

(ii) State the modal fuel efficiency class and calculate its mode. [1]

(iii) What percentage of trucks achieved an optimal mileage of 16 km/l or higher? [1]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Subpart (i): Arithmetic Mean Fuel Efficiency [2 Marks]

Mileage (km/l)	Trucks (f_i)	Class Mark (x_i)	$f_i x_i$
$10-12$	7	11	77
$12-14$	12	13	156
$14-16$	18	15	270
$16-18$	13	17	221
$18-20$	10	19	190
Total	$\sum f_i = 60$		$\sum f_i x_i = 914$

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{914}{60} = \frac{457}{30} \approx \mathbf{15.23} \text{ km/l} \quad \left(\text{or } 15\frac{7}{30} \text{ km/l}\right).$$

$$\boxed{\text{Mean Fuel Efficiency} = \mathbf{15.23} \text{ km/l} \quad \left(\text{or } \mathbf{15\frac{7}{30}} \text{ km/l}\right)}$$

Subpart (ii): Modal Class & Mode Calculation [1 Mark]

The highest frequency is 18 $\implies$ Modal Class = **14 – 16** km/l.
$l = 14, h = 2, f_1 = 18, f_0 = 12, f_2 = 13$.

$$\text{Mode} = 14 + \left(\frac{18-12}{2(18)-12-13}\right) \times 2 = 14 + \left(\frac{6}{11}\right) \times 2 = 14 + \frac{12}{11} \approx \mathbf{15.09} \text{ km/l}.$$

$$\boxed{\text{Modal Class} = \mathbf{14-16} \text{ km/l}, \quad \text{Mode} = \mathbf{15.09} \text{ km/l}}$$

Subpart (iii): Percentage Achieving ≥ 16 km/l [1 Mark]

Trucks in intervals $[16-18]$ and $[18-20] = 13 + 10 = 23$ trucks.

$$\text{Percentage} = \frac{23}{60} \times 100\% = \frac{230}{6}\% = \mathbf{38.33\%} \quad \left(\text{or } 38\frac{1}{3}\%\right).$$

$$\boxed{\text{Percentage of Trucks} \geq 16 \text{ km/l} = \mathbf{38.33\%}}$$

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